v44 model — experimental conditions: input parameters & where they sit in the proliferation parameter space

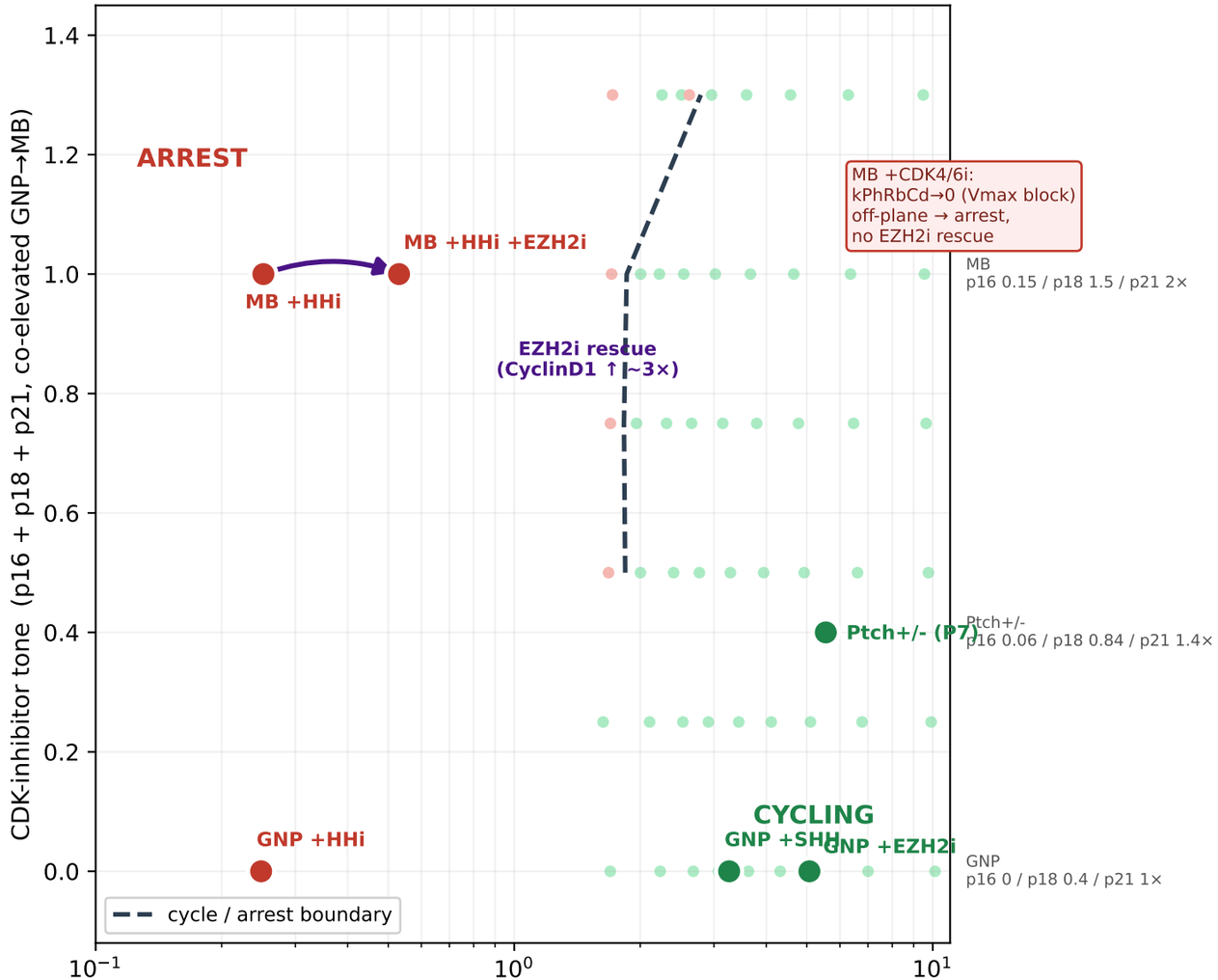
A Input parameters per condition

	SHH	Ptch1	fMYCN	p16	p18	p21	HHi	EZH2i	HU	CDK4/6i	cyc?
GNP +SHH	0.5	1	1	0	**0.4**	1×	·	·	·	·	●
GNP +HHi	0.5	1	1	0	**0.4**	1×	✓	·	·	·	○
GNP +EZH2i	0.5	1	1	0	**0.4**	1×	·	✓	·	·	●
GNP serum-starve	0.5	1	1	0	**0.4**	1×	·	·	·	·	○
Ptch+/- (P7)	0.5	**0.5**	**1.38**	**0.06**	**0.84**	**1.4×**	·	·	·	·	●
MB	0.5	**0.1**	**2.86**	**0.15**	**1.5**	**2×**	·	·	·	·	●
MB +HHi	0.5	**0.1**	**2.86**	**0.15**	**1.5**	**2×**	✓	·	·	·	○
MB +HHi +EZH2i	0.5	**0.1**	**2.86**	**0.15**	**1.5**	**2×**	✓	✓	·	·	○
MB +CDK4/6i	0.5	**0.1**	**2.86**	**0.15**	**1.5**	**2×**	·	·	·	✓	○
MB +HU	0.5	**0.1**	**2.86**	**0.15**	**1.5**	**2×**	·	·	✓	·	●

context inputs (p16, p18, p21 = CDK-inhibitor tone) *drug switches*

p16: competitive CDK4/6 brake (raises K_CdRb) · p21: CDK2 inhibitor (kSyP21, ×baseline) · serum-starve k_Cd_trans→0 · CDK4/6i kPhRbCd→0

B Parameter space: CyclinD1 vs the CDK-inhibitor brake



points = deterministic mean cell · tumour heterogeneity straddles the boundary → fractional response (MB+HHi 20% → +EZH2i 74% cycling)